COURSE CODE:22PHY12B

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RV COLLEGE OF ENGINEERING
Autonomous Institution affiliated to VTU
I Semester B.E. April -2023 Examinations
DEPARTMENT: PHYSICS.
COURSE TITLE: CLASSICAL PHYSICS FOR ENGINEERS
(2022 SCHEME)
(Integrated Course – Lab + Theory)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10, and 11 lab components (compulsory).

PART-A (Objective type for one or two marks)
(True & false and match the following questions are not permitted)

1	1.1	What is neutral surface of a cantilever?	10
	1.2	Why does it feel difficult to twist a short stub wire?	
	1.3	Define sharpness of resonance.	
	1.4	A damped oscillator undergoes periodic motion. What is the type of damping it is experiencing?	
	1.5	State the zeroth law of thermodynamics.	
	1.6	State the law of Adiabatic expansion of a gas.	
	1.7	Define contact angle.	
	1.8	State an application of Bernoulli's principle.	
	1.9	What is magnetic particle testing used for ?	
	1.10	State one advantage of TEM over SEM.	

PART-B (Maximum subdivisions is limited to 2 in each question)

		UNIT-I	
2	a	Set up the differential equation for a damped harmonic oscillator and obtain the expression for displacement. Discuss the case of under damping. Starting with the expressions for the amplitude and phase of a forced, damped harmonic oscillator, work out the specific cases for a) driving frequency much greater than natural frequency, b) Resonance	14
	b	A body of mass m is attached by two springs of spring constant k_1 and k_2 in series to a fixed support. If $k_1 = 2k_2$ and the external mass attached is $6\pi^2/k_2$, calculate the time period of the resultant oscillation.	

		UNIT-II	
3	a	Starting from the moment balance equation, derive an expression for the	

		end point depression of a cantilever of length L and circular cross section r and Young's modulus Y. With the help of a diagram show that shearing strain is equal to the sum of tensile strain and compressive strain.	14
	b	Find the angular twist of a wire of length 0.3m, and radius $0.2 \times 10^{-3}m$ when a torque of $5 \times 10^{-4}Nm$ is applied. Rigidity modulus of the material $8 \times 10^{10}Nm^2$.	
4	a	Derive an expression for the couple per unit twist of a wire. With the help of a neat diagram, explain the stress-strain relationship of an elastic body subjected to an external load	
	b	A wire of radius 4 mm and length 20 cm undergoes an elongation of 2 mm when subjected to an external load of 100 g. Calculate the Young's modulus and the rigidity modulus, given that Poisson's ratio = 0.4	

		UNIT-III	
5	a	With a diagram, explain the zeroth law of thermodynamics. Derive an expression for the work done in an adiabatic process.	14
	b	Calculate the work done by a gas undergoing isothermal expansion at T = 1000 K from initial volume = 100 cc to final volume 278 cc.	
		OR	
6	a	With a neat diagram explain Joule's experiment. Derive an expression for the work done in an isothermal process.	
	b	Calculate the work done by a gas undergoing adiabatic expansion from initial volume of 100 cc to final volume of 116 cc as the pressure is reduced from 100 kilo-bar to 80 kilo-bar.	

UNIT-IV			
7	a	With the help of a diagram, distinguish between Newtonian and non-Newtonian fluids. Derive an expression for the capillary rise of a liquid in a narrow glass cylinder immersed in it.	14
	b	A 0.6 mm diameter glass tube is inserted into water at 20°C in a cup. Determine the capillary rise of water in the tube.	
		OR	
8	a	Derive the continuity equation of fluid flow in integral form.	
	b	Show that the bulk modulus of an ideal gas is equal to its absolute pressure.	

UNIT-V			
9	a	With a neat labelled diagram explain the principle, construction and working of X-ray diffractometer.	14
	b	Discuss two types of destructive and non-destructive testing.	
		OR	
10	a	With a neat labelled diagram explain the principle, construction and working of Atomic force microscope.	
	b	Explain the different types of emissions that happen from the sample in a scanning electron microscope.	

LAB COMPONENT

11 a — Draw the circuit diagram of the experiment done to calculate the Fermi energy of Copper. The data collected for a Fermi energy experiment is given in the table below. Calculate the slope and the intercept and thereby the Femi energy of the sample. — 20

Sl. No.	Temp in 0C	Temp in Kelvin	R in (Ohm)
Room Temp	19.4	292.4	9.9
01	80	353.15	13
02	75	348.15	12.8
03	70	343.15	12.6
04	65	338.15	12.4
05	60	333.15	12.2

06	55	328.15	12
07	50	323.15	11.8
08	45	318.15	11.6
09	40	313.15	11.4
10	35	308.15	11.2